F79SP Session 1: Introduction to Stochastic Processes

1 Revision on analysis

Not all infinities are the same. A set S with cardinality $|S|$ is called:

- finite if $|S| < |\mathbb{N}|$.
- countably infinite if $|S| = |\mathbb{N}|$.
- countable if it is either finite or countably infinite.
- uncountable if $|S| > |\mathbb{N}|$.

Remarks:

- Informally, countably infinite can be regarded as the "smallest" infinity.
- Some examples of countable sets are $\mathbb{N}, \mathbb{Z}$ and $\mathbb{Q}$, and examples of uncountable sets are $[0, 1], [1, 2] \cup [3, 4]$ and $\mathbb{R}$.

2 Stochastic processes

A stochastic process is a collection of random variables $\{X_t\}_{t \in T}$ ordered in time $t \in T$. A classification of stochastic processes:

- Disrete-time: Index set T is countable.
- Continuous-time: Index set T is a connected subset of $\mathbb{R}$.

In general, $t = 0$, $n = 0$ means now, $t > 0$, $n > 0$ are times in futures. In practice, we normally use the following T:

- Discrete-time: $T =$ and we write $\{X_n\}_{n=0}^{\infty}$ or $\{X_n\}_{n \geq 0}$.
- Continuous-time: $T =$ and we write $\{X_t\}_{t \geq 0}$.

The underlying random variables can be discrete, continuous r.v.'s etc.

Example 2.0.1 (Bernoulli Processes)**.** *A discrete-time stochastic process* $\{X_n\}_{n\geq 0}$ *with i.i.d.* $X_i \sim$ *Bernoulli*(p) *is called a Bernoulli process.*

- *At each time step* i*,* X_i *takes either* 1 *or* 0 *as values, with probability* p *and* $1-p$ *respectively.*

Some further examples of stochastic processes are as below.

1. X_n is insurance premium paid in nth year of policy by some policy holder.
2. X_t is value of share at time t.
3. X_t is state of person's being at time t, i.e., healthy, sick or dead.

3 Limit theorems

We first consider three types of convergence.

Types of convergence

Let $X_1, X_2, \ldots$ be a sequence of r.v.'s and X be another r.v.'s. We say $\{X_n\}_{n\geq 0}$ converges to X:

- in distribution ($X_n \to_D X$) if $\lim_{n\to\infty} F_n(x) = F(x)$ for (almost all) x.
- in probability ($X_n \to_P X$) if for all $a > 0$, $\lim_{n\to\infty} \mathbf{P}\left(\left|X_n - X\right| > a\right) = 0$.
- almost surely ($X_n \to_{a.s.} X$) if $\mathbf{P}(X_n \to X) = 1$.

(Weak) Law of Large numbers (WLLN or LLN)

Theorem 3.0.1. *Let $X_1, X_2, \ldots$ be a sequence of i.i.d. r.v.'s.*

1. *Mean $\mu = \mathbf{E}X_1$ and variance $\sigma^2 = \mathbf{Var}X_1$ are finite.*
2. *Let $S_n = X_1 + X_2 + \cdots + X_n$.*
3. *Fix any $a > 0$.*

Then, when $n \to \infty$, we have:

$$\lim_{n\to\infty} \mathbf{P}\left(\left|\frac{S_n}{n} - \mu\right| > a\right) = 0.$$

or in short, $S_n/n \to_P \mu$.

Proof. Refer to lecture notes. The proof follows from Chebyshev inequality, which says for any r.v. Y with $\mathbf{E}Y = \mu$, $\mathbf{Var}Y < \infty$. Fix any $a > 0$. Then:

$$\mathbf{P}\left(\left|Y - \mu\right| > a\right) \leq \frac{\mathbf{Var}Y}{a^2}.$$

Let $Y = S_n/n$. We have:

- $\mathbf{E}Y = \mathbf{E}(S_n/n) =$
- $\mathbf{Var}Y =$
- Substitute into Chebyshev inequality:

$$\mathbf{P}\left(\left|\frac{S_n}{n} - \mu\right| > a\right) \leq \frac{\sigma^2}{na^2}.$$

- Take limit $n \to \infty$ on both sides yield the result.

□

Remarks:

1. The sequence $\mathbf{P}\left(\left|\frac{S_1}{1} - \mu\right| > a\right)$,$\mathbf{P}\left(\left|\frac{S_2}{2} - \mu\right| > a\right), \ldots, \mathbf{P}\left(\left|\frac{S_n}{n} - \mu\right| > a\right), \ldots$ converges to value 0.
2. Informally, we say the average of observation $\frac{S_n}{n}$ will be closed to the expected value μ in a high probability (within any fixed margin $a > 0$).
3. This **is not the same as** $\{\frac{S_n}{n} - \mu\}_{n\geq 0}$ converges to 0 in value.

Strong Law of Large Numbers (SLLN)

Theorem 3.0.2. *Let $X_1, X_2, \ldots$ be a sequence of i.i.d. r.v.'s.*

1. *Mean $\mu = \mathbf{E}X_1$ and variance $\sigma^2 = \mathbf{Var}X_1$ are finite.*
2. *Let $S_n = X_1 + X_2 + \cdots + X_n$.*

Then, when $n \to \infty$, we have:

$$\mathbf{P}\left(\left|\frac{S_n}{n} - \mu\right| \to 0\right) = 1.$$

or in short, $\frac{S_n}{n} \to_{a.s} \mu$.

Remarks: We can interpret $\left|\frac{S_n}{n} - \mu\right| \to 0$ as follows. By definition of limits, for every $\epsilon > 0$, there exists $N \in \mathbb{Z}^+$ such that for $n > N$, $\left|\left|\frac{S_n}{n} - \mu\right| - 0\right| = \left|\frac{S_n}{n} - \mu\right| < \epsilon$. Diagram-wise:

Some general properties (without proof):

1. $X_n \to_{a.s.} X \Rightarrow X_n \to_P X \Rightarrow X_n \to_D X$.
2. Let c be a constant. $X_n \to_D c \Leftrightarrow X_n \to_P c$.